$V_A - V_B$

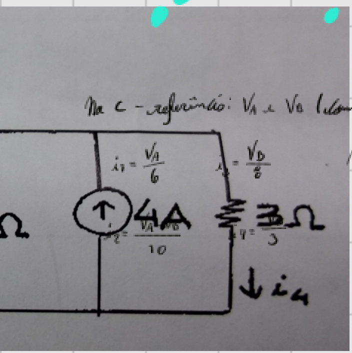
A B

i_1 i_3

V_A V_B V_B V_B

C

	Tensão	Corrente	Potência	
$R_1 = 6\Omega$	27V	4,5A	121,5W	210
$R_2 = 10\Omega$	15V	1,5A	22,5W	
$R_3 = 8\Omega$	12V	1,5A	18W	
$R_4 = 3\Omega$	12V	4A	48W	
fonte 6A	27V	6A	162W	210
fonte 4A	12V	4A	48W	



Nó C - referência: V_A e V_B (com relação a referência)

$i_1 = \frac{V_A}{6}$ $i_3 = \frac{V_B}{8}$ $i_2 = \frac{V_A - V_B}{10}$ $i_4 = \frac{V_B}{3}$

Nó A: $i_1 + i_2 = 6$

$\frac{V_A}{6} + \frac{V_A - V_B}{10} = 6$

$\boxed{8V_A - 3V_B = 180}$

Nó B: $i_2 + 4 = i_3 + i_4$

$\frac{V_A - V_B}{10} + 4 = \frac{V_B}{8} + \frac{V_A}{3}$

$\boxed{-12V_A + 67V_B = 480}$

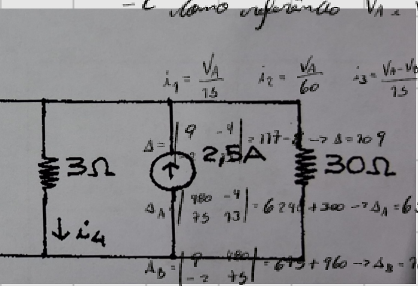
$$\begin{cases} 8V_A - 3V_B = 180 \xrightarrow{3x} 24V_A - 9V_B = 540 \\ -12V_A + 67V_B = 480 \xrightarrow{2x} -24V_A + 134V_B = 960 \end{cases}$$

$\boxed{V_B = \frac{1500}{125} = 12} \to \boxed{V_A = 27}$

A B

i_1 i_2 i_5 i_8

C



- C como referência V_A e V_B $V = R.i$ $i = \frac{V}{R}$

$i_1 = \frac{V_A}{15}$ $i_2 = \frac{V_A}{60}$ $i_3 = \frac{V_A - V_B}{15}$ $i_4 = \frac{V_B}{3}$ $i_5 = \frac{V_B}{30}$

$\Delta = \begin{vmatrix} 9 & -4 \\ -2 & 13 \end{vmatrix} = 117 - 8 \to \Delta = 109$

$\Delta_A = \begin{vmatrix} 480 & -4 \\ 75 & 13 \end{vmatrix} = 6240 + 300 \to \Delta_A = 6540$

$\Delta_B = \begin{vmatrix} 9 & 480 \\ -2 & 75 \end{vmatrix} = 675 + 960 \to \Delta_B = 1635$

$V_A = \frac{\Delta A}{\Delta} = \frac{6540}{109} \to V_A = 60A$ $V_B = \frac{\Delta B}{\Delta} = \frac{1635}{109} \to V_B = 15V$

* Nó A:

$i_1 + i_2 + i_3 = 8$

$\boxed{9V_A - 4V_B = 480}$

* Nó B:

$i_4 + i_5 = i_3 + 2,5$

$\boxed{-2V_A + 13V_B = 75}$

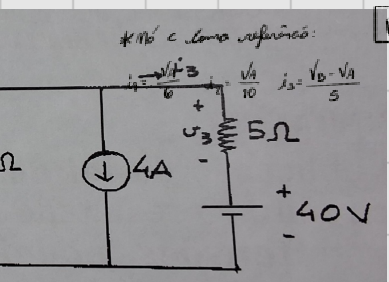
	Tensão	Corrente	
$R_1 = 15$	60	4	517,5W
$R_2 = 60$	60	1	
$R_3 = 15$	45	3	
$R_4 = 3$	15	5	
$R_5 = 30$	15	0,5	
fonte 8A	60	8	517,5W
fonte 2,5A	15	2,5	

A

B i_5

i_2

C



* Nó C como referência:

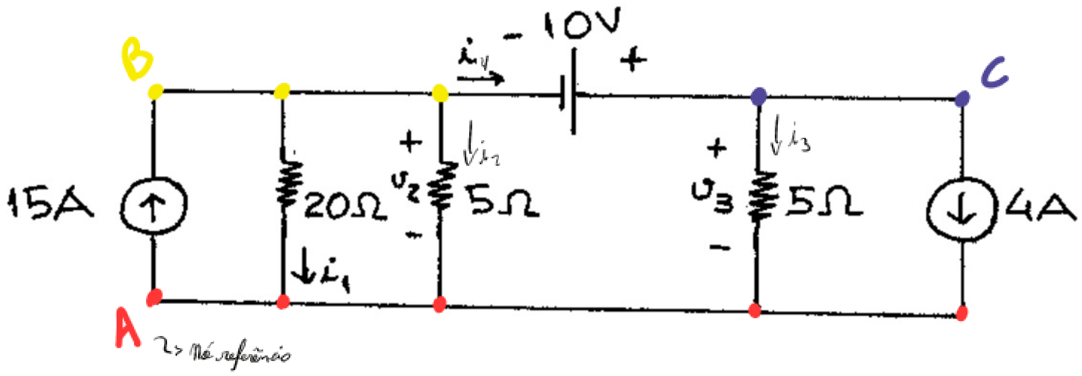
$i_1 = \frac{V_A + 3}{6}$ $i_2 = \frac{V_A}{10}$ $i_3 = \frac{V_B - V_A}{5}$

$\boxed{V_B = 40}$

* Nó A: $10 + i_3 = i_1 + i_2 + 4$

$5V_A + 3V_A = 6V_B - 6V_A + 180 \to \boxed{V_A = 30V}$

4)



$V = R \cdot i \rightarrow i = \frac{V}{R}$

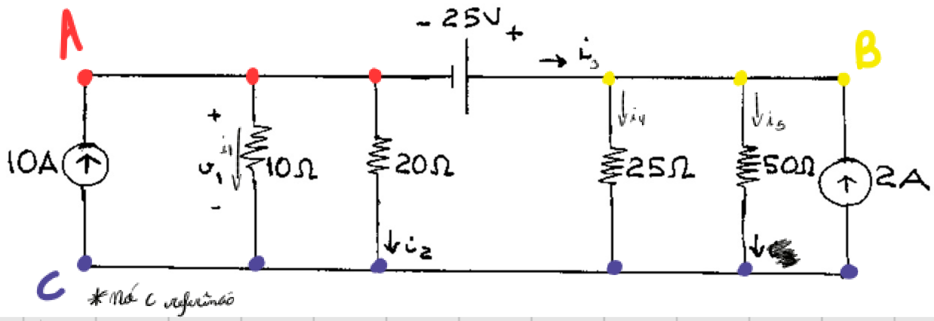
$i_1 = \frac{V_B}{20}$ $\quad i_2 = \frac{V_B}{5}$ $\quad i_3 = \frac{V_C}{5}$ $\quad V_C - V_B = 10$

∗ Nó B: $i_1 + i_2 + i_4 = 15 \rightarrow i_1 + i_2 + i_3 + 4 = 15$

∗ Nó C: $i_4 = i_3 + 4$ $\quad \frac{V_B}{20} + \frac{V_B}{5} + \frac{V_C}{5} = 11$ (I)

(I) $\frac{V_B}{20} + \frac{V_B}{5} + \frac{V_C}{5} = 11 \xrightarrow{\times 20} V_B + 4V_B + 4 \cdot (V_B + 10) = 220 \rightarrow 9V_B = 180 \rightarrow \boxed{V_B = 20V}$ ∴ $\boxed{V_C = 30V}$

5)



$V = R \cdot i \rightarrow i = \frac{V}{R}$

$i_1 = \frac{V_A}{10}$ $\quad i_2 = \frac{V_A}{20}$ $\quad i_4 = \frac{V_B}{25}$ $\quad i_5 = \frac{V_B}{50}$

$V_B - V_A = 25$

$V_B = 25 + V_A$

∗ Nó A: $i_1 + i_2 + i_3 = 10 \rightarrow i_1 + i_2 + i_4 + i_5 = 12$

∗ Nó B: $i_4 + i_5 = i_3 + 2$

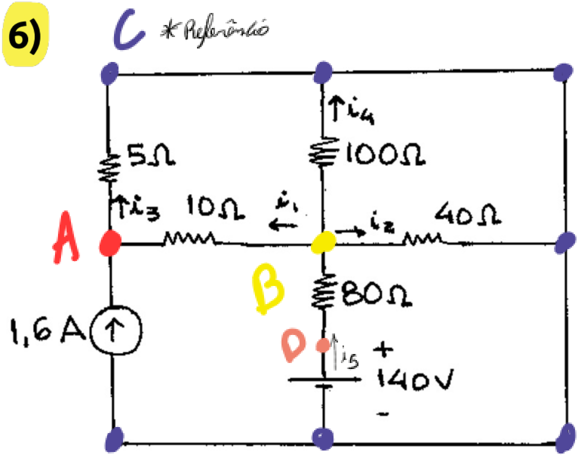
$\boxed{i_3 = i_4 + i_5 - 2}$

$\frac{V_A}{10} + \frac{V_A}{20} + \frac{V_B}{25} + \frac{V_B}{50} = 12 \rightarrow 15V_A + 6V_B = 1200$

$15V_A + 6 \cdot (V_A + 25) = 1200$

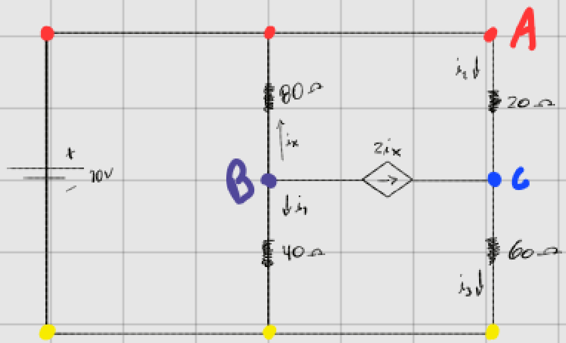
$21V_A = 1050 \rightarrow V_A = 50V$ ∴ $V_B = 75V$

6)



$V = R \cdot i \rightarrow i = \frac{V}{R}$ $\quad \boxed{V_D = 140V}$

7 Circuito com fonte dependente:



D → Referência

$i_1 = \frac{V_B}{40}$ $i_2 = \frac{V_A - V_C}{20}$ $i_3 = \frac{V_C}{60}$ $i_x = \frac{V_B - V_A}{80}$

* Nó A: $V_A = 10V$

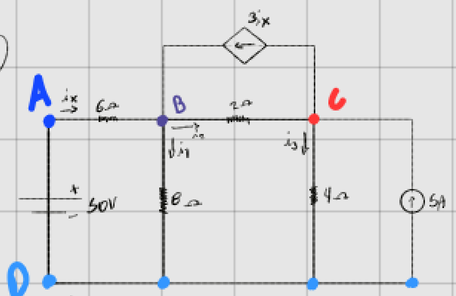
* Nó B:

$i_x + 2i_x + i_1 = 0 \rightarrow \frac{3(V_B - V_A)}{80} + \frac{V_B}{40} = 0 \rightarrow V_B = 6V$

* Nó C:

$2i_x + i_2 = i_3 \rightarrow \frac{2.(V_B - V_A)}{80} + \frac{(V_A - V_C)}{20} = \frac{V_C}{60} \rightarrow 8V_C = 3V_B + 3V_A \rightarrow V_C = 6V$

8



↳ Referência

* Nó A: $V_A = 50V$

* Nó B:

$i_1 + i_2 = i_x + 3i_x \rightarrow \frac{(V_B - V_C)}{2} + \frac{V_B}{8} = \frac{4.(V_A - V_B)}{6} \rightarrow 31V_B - 12V_C = 800$

* Nó C:

$3i_x + i_3 = i_2 + 5 \rightarrow \frac{3.(V_A - V_B)}{6} + \frac{V_C}{4} = \frac{V_B - V_C}{2} + 5 \rightarrow 4V_B - 3V_C = 2V_A + 20 \rightarrow 4V_B - 3V_C = 80$

$$\begin{cases} 31V_B - 12V_C = 800 \\ 4V_B - 3V_C = 80 \end{cases} \rightarrow \begin{matrix} V_B = 32V \\ V_C = 16V \end{matrix} \quad e \quad V_A = 10V$$